

Python and Ruby. This AI framework, which comes with over 80+ world languages, can easily be integrated into mobile apps, websites, Facebook messenger, Slack, smart wearable devices, and home automation. When these frameworks are developed into symptom checkers, they can efficiently combine rule-based algorithms with natural language processing (NLP) libraries to render the most relevant medical prognosis.

Neural networks for disease progression modelling

Neural networks, a form of deep learning computational model, can be trained to perform tasks like classification, prediction, visualisation, and decision-making. Artificial neural networks, also called artificial neurons and organised in different interconnected layers called input, hidden, and output layers, work on feed-forward and/or back-propagation algorithms to mass classify data sets

and generate desired outputs. While there are different types of artificial neural networks, deep convolutional neural networks (DCNNs) and recurrent neural networks (RNNs) have the utmost relevance in the healthcare domain.

DCNNs use many hidden layers to perform convolutions and sub-sampling of the raw input data. The hidden layers process repetitive convolutions using unsupervised classification methodologies until features to the most granular pixel levels are extracted, and the estimated class is computed for the output layer. DCNN also uses ML methods like back-propagation techniques and the rescaling of gradients to minimise classification errors on the output data. In medical imaging and chest x-rays, the DCNN feature extraction process can better detect disease pathology and monitor progression. The data augmentation of DCNN is 96.6 per cent accurate.

RNNs in medical diagnoses are used for analysing temporal data sets.

Neurons like nodes arranged in successive layers work on one-way feed-forward propagation mechanisms. RNNs thrive on supervised classification. Sample training data sets, along with the raw input data, are required to perform feature extraction and image classifications. Although the accuracy rate of prognosis in RNNs is a little lesser than that in DCNNs, it is still preferred by medical data analysts for its longitudinal analysis capabilities. RNNs can read medical histories, analyse temporal aspects of the illness, infer the current state of the pathology, and predict future disease progressions. Multi-scale temporal pooling of disease data helps the network monitor the disease progression more holistically.

AI-based neural networks let professionals optimise their work and time better. Neural networks help the medical fraternity deliver better healthcare services during time-critical

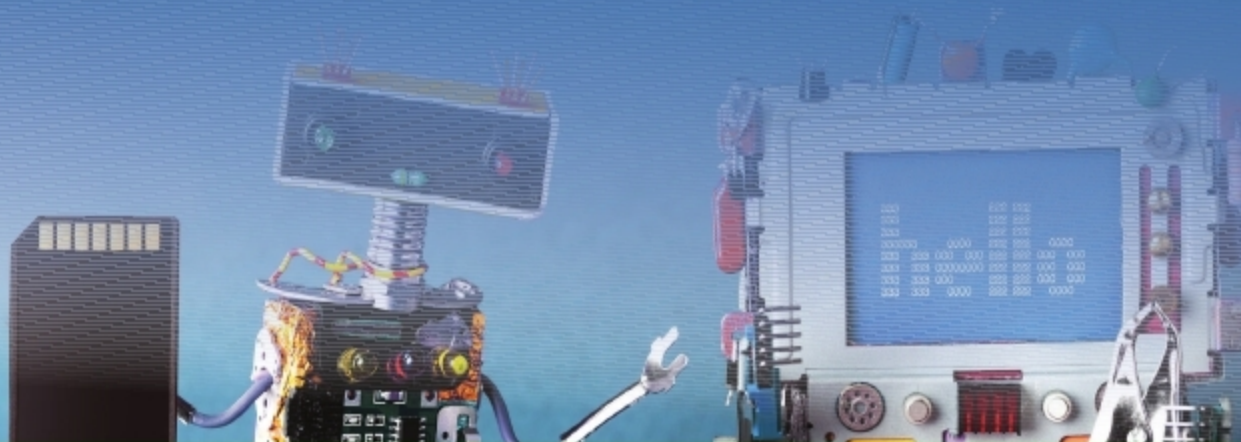
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